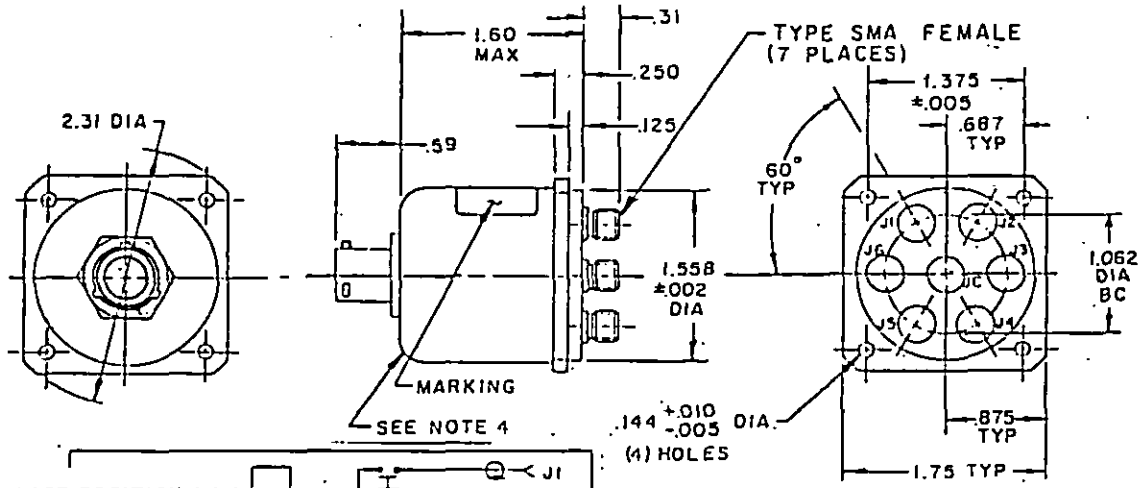
MILITARY SPECIFICATION SHEET

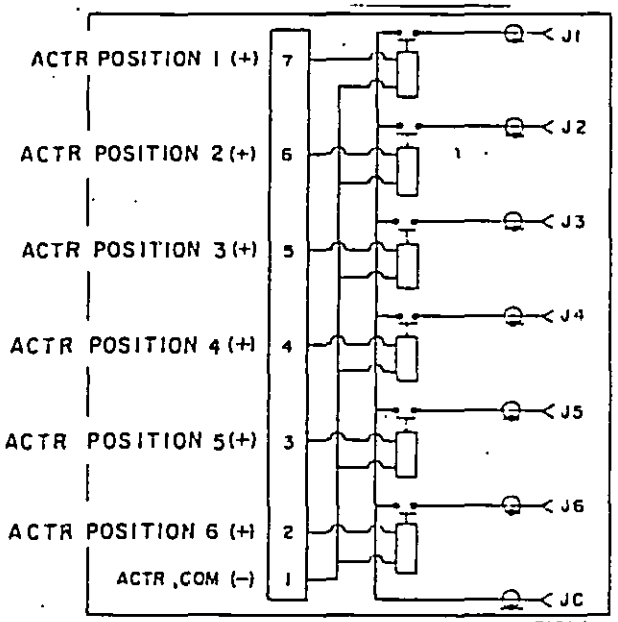
SWITCHES, RADIO FREQUENCY TRANSMISSION LINE (COAXIAL) (ELECTRICALLY OPERATED) CLASS 5, 1P6T

(B)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the switches described herein shall consist of this specification and the latest issue of MIL-S-3928.





SCHEMATIC (SHOWN IN DEENERGIZED POSITION)

Inches	mm
.002	0.05
.005	0.13
.010	0.25
.125	3.18
.144	3.66
.25	6.4
.31	7.9
.59	15.0
.687	17.45
.875	22.22
1.062	26.97
1.375	34.92
1.558	39.57
1.60	40.6
1.75	44.4
2.31	58.7

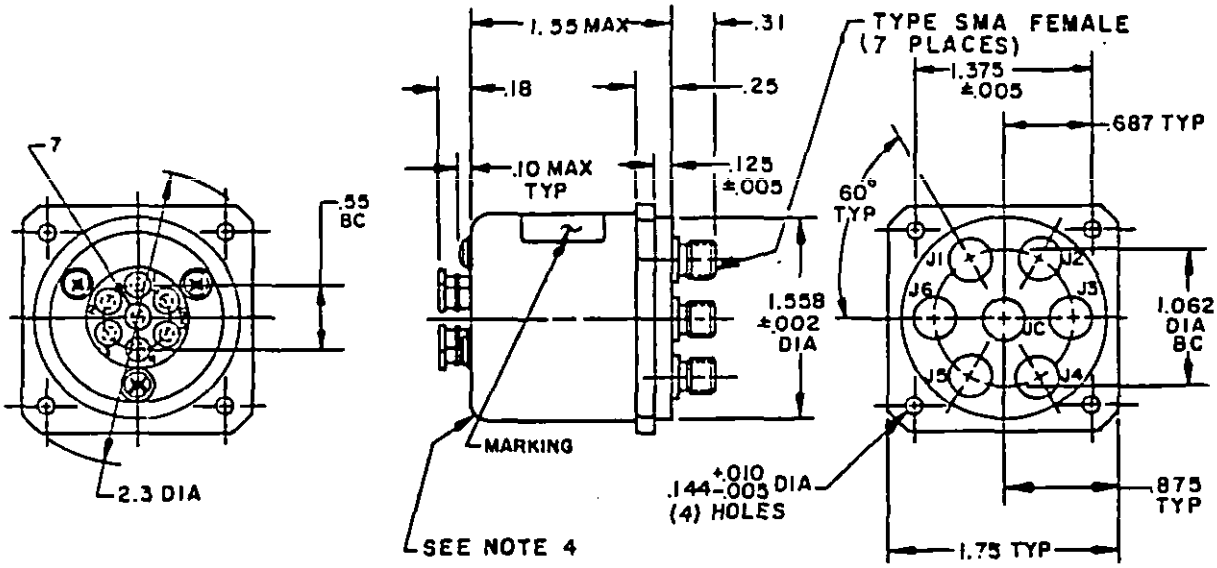
NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.010 (0.25 mm) for three place decimals and ±0.03 (0.8 mm) for two place decimals.
4. Round corners of case may be squared.

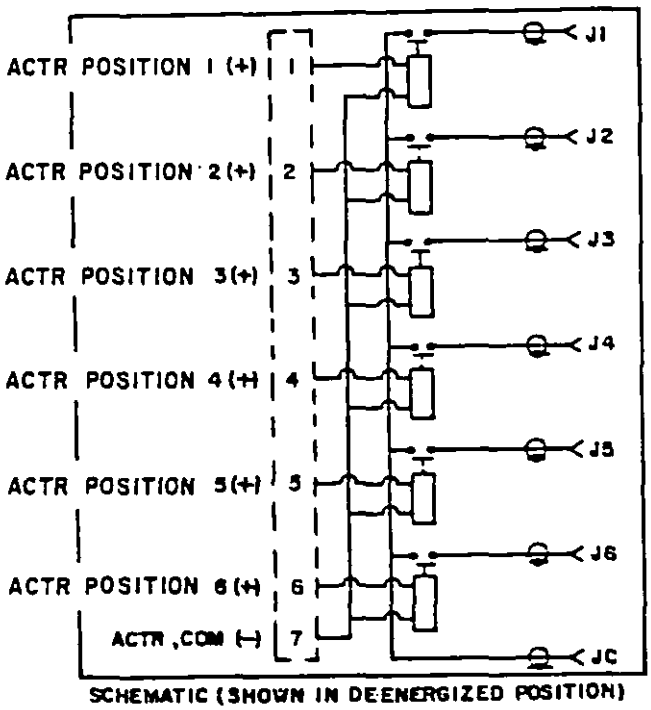
FIGURE 1. Switch configuration and schematic, part number M3928/18-01.

(B) denotes changes

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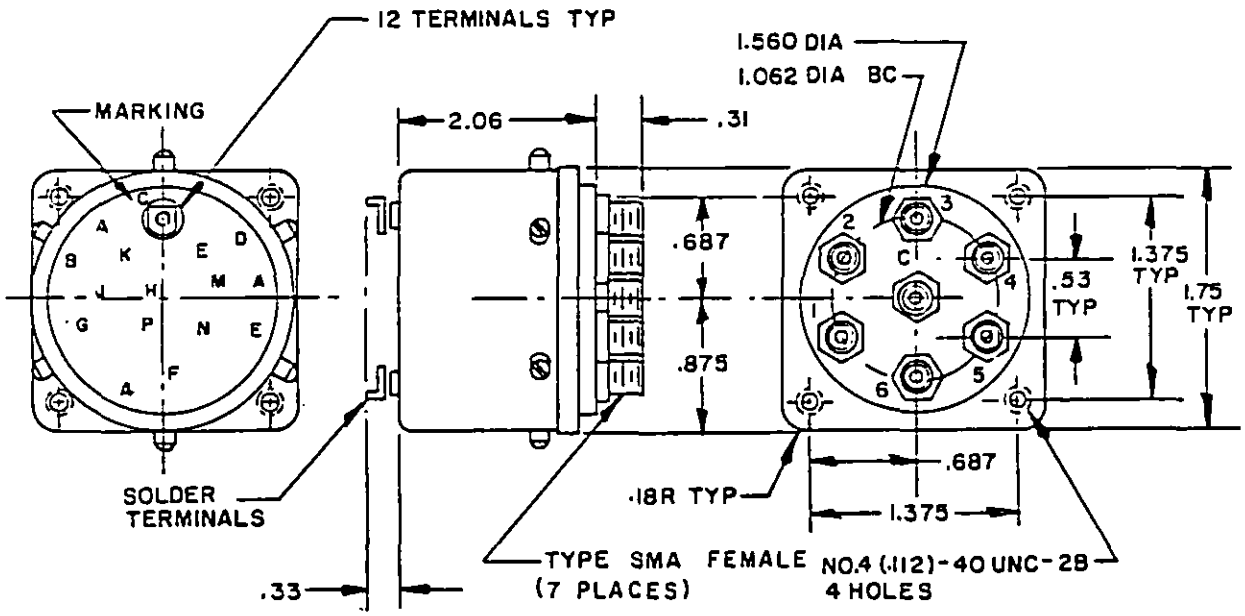
SCHEMATIC (SHOWN IN DEENERGIZED POSITION)

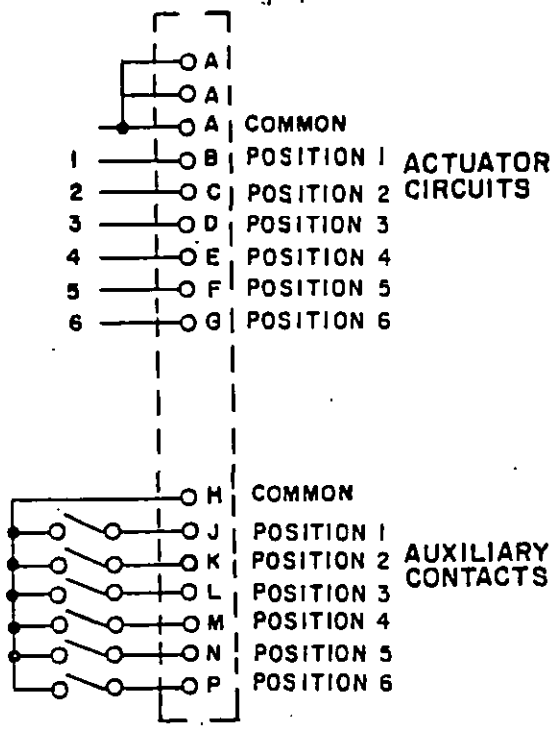
Inches	mm
.002	0.05
.005	0.13
.010	0.25
.125	3.18
.144	3.66
.25	6.4
.31	7.9
.59	15.0
.687	17.45
.875	22.22
1.062	26.97
1.375	34.92
1.558	39.57
1.60	40.6
1.75	44.4
2.31	58.7

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.010 (0.25 mm) for three place decimals and ±0.03 (0.8 mm) for two place decimals.
4. Round corners of case may be squared.

(B) FIGURE 2. Switch configuration and schematic, part numbers M3928/18-02 and M3928/18-08.

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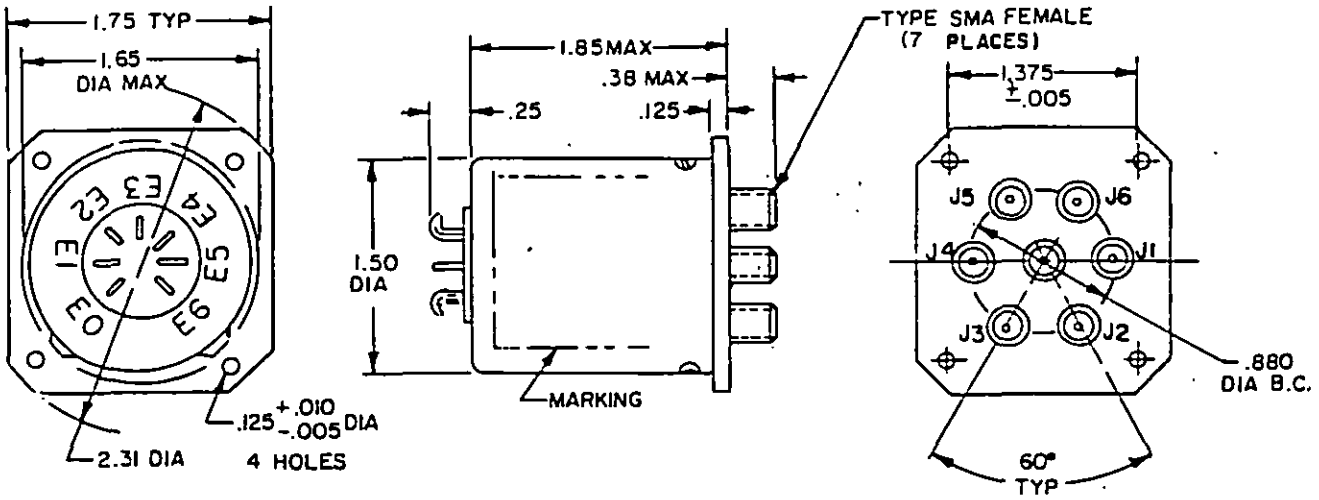
Inches	mm
.112	2.84
.31	7.9
.33	8.4
.687	17.45
.875	22.22
1.062	26.97
1.375	34.92
1.560	39.62
1.75	44.4
2.06	52.3

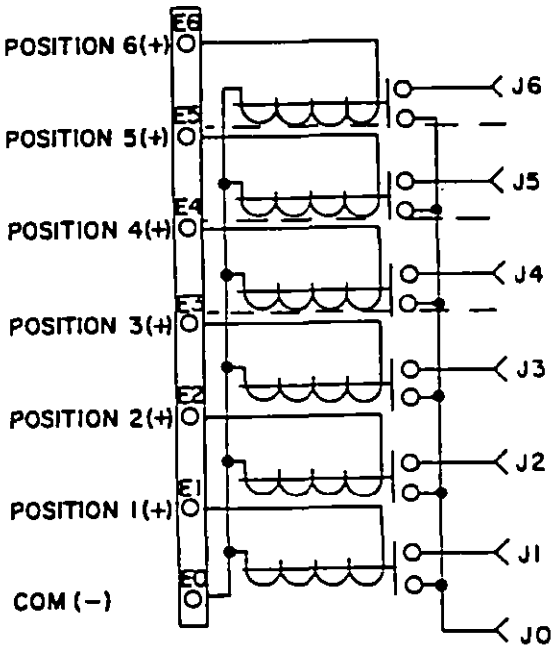
SCHEMATIC

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Metric equivalents are in parentheses.
4. Round corners of case may be squared.
5. Unless otherwise specified, tolerances are ±.005 (0.13 mm) for three place decimals and ±.02 (0.5 mm) for two place decimals.

FIGURE 3. Outline and schematic drawing, part number M3928/18-03.

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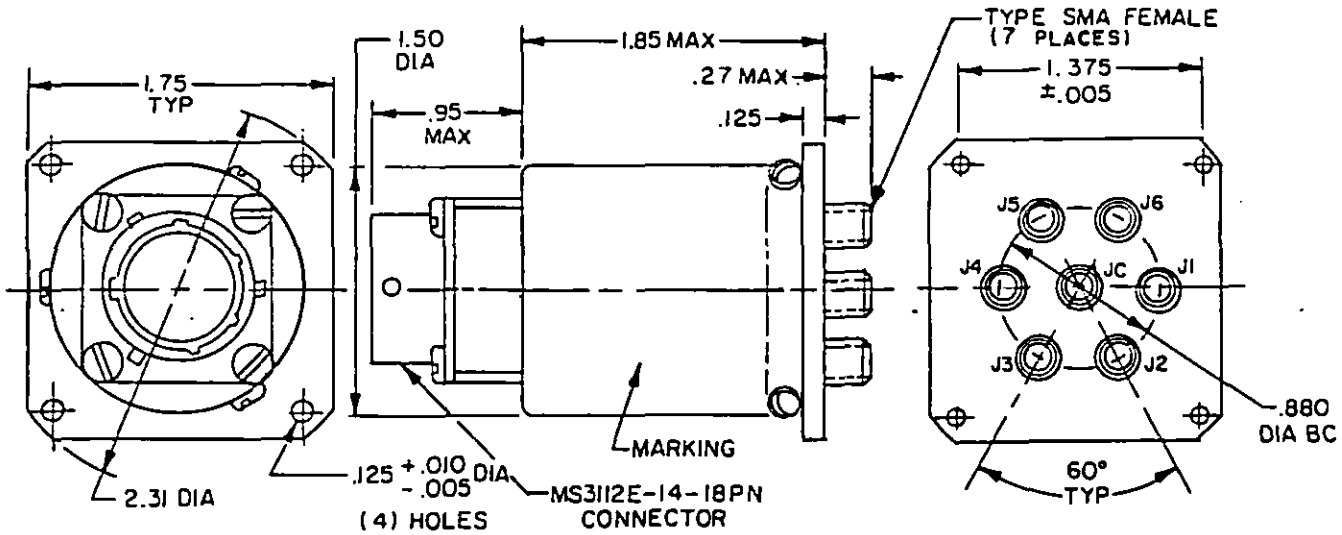
SCHEMATIC (SHOWN IN DEENERGIZED POSITION)

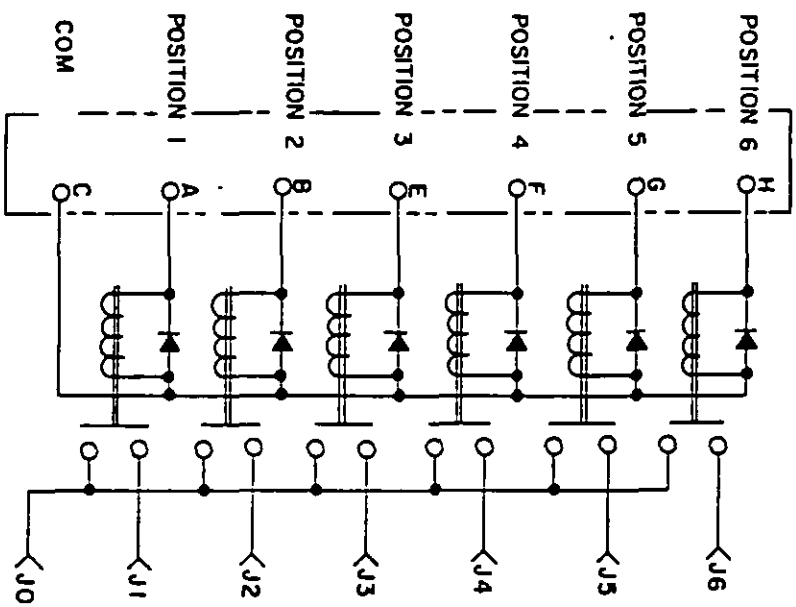
Inches	mm
.005	0.13
.125	3.18
.25	6.4
.38	9.7
.880	22.35
1.375	34.93
1.50	38.1
1.65	41.9
1.75	44.5
1.85	47.0
2.31	58.7

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.005 (0.13 mm) for three place decimals and ±.02 (0.5 mm) for two place decimals.

(8) FIGURE 4. Switch configuration and schematic, part number M3928/18-04.

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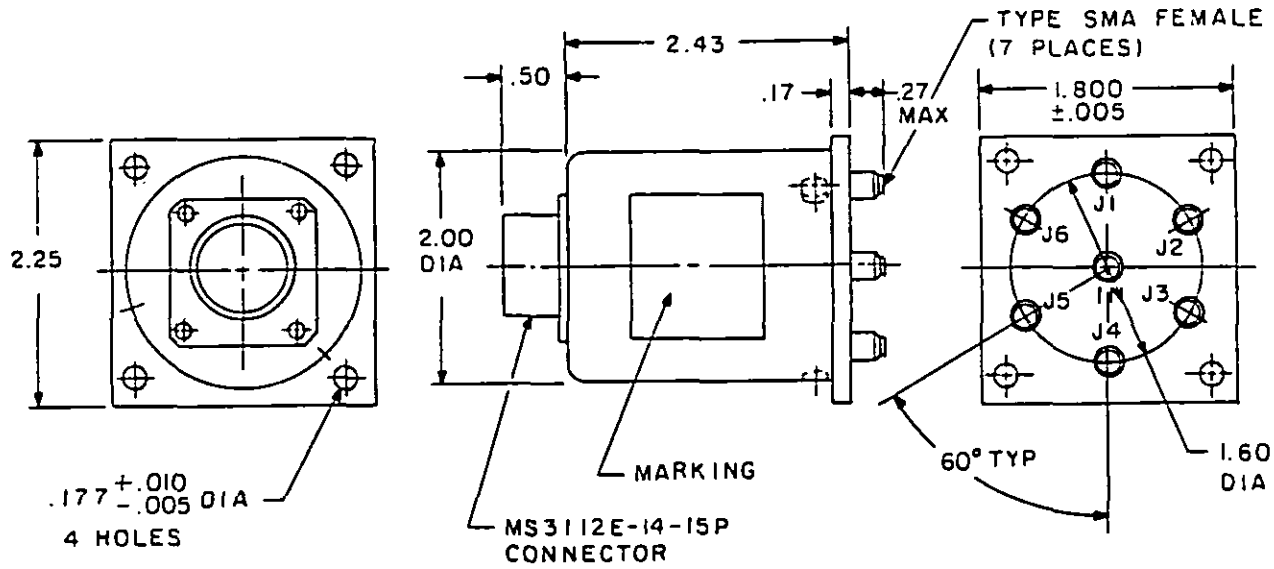
Inches	mm
.005	0.13
.125	3.18
.25	6.4
.38	9.7
.880	22.35
.95	24.1
1.375	34.93
1.50	38.1
1.65	41.9
1.75	44.5
1.85	47.0
2.31	58.7

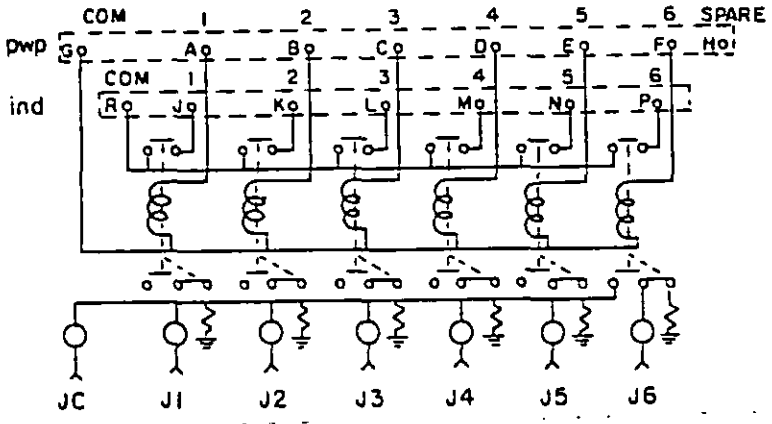
SCHEMATIC (SHOWN IN DEENERGIZED POSITION)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.005 (0.13 mm) for three place decimals and ±.02 (0.5 mm) for two place decimals.

Ⓑ FIGURE 5. Switch configuration and schematic, part number M3928/18-05.

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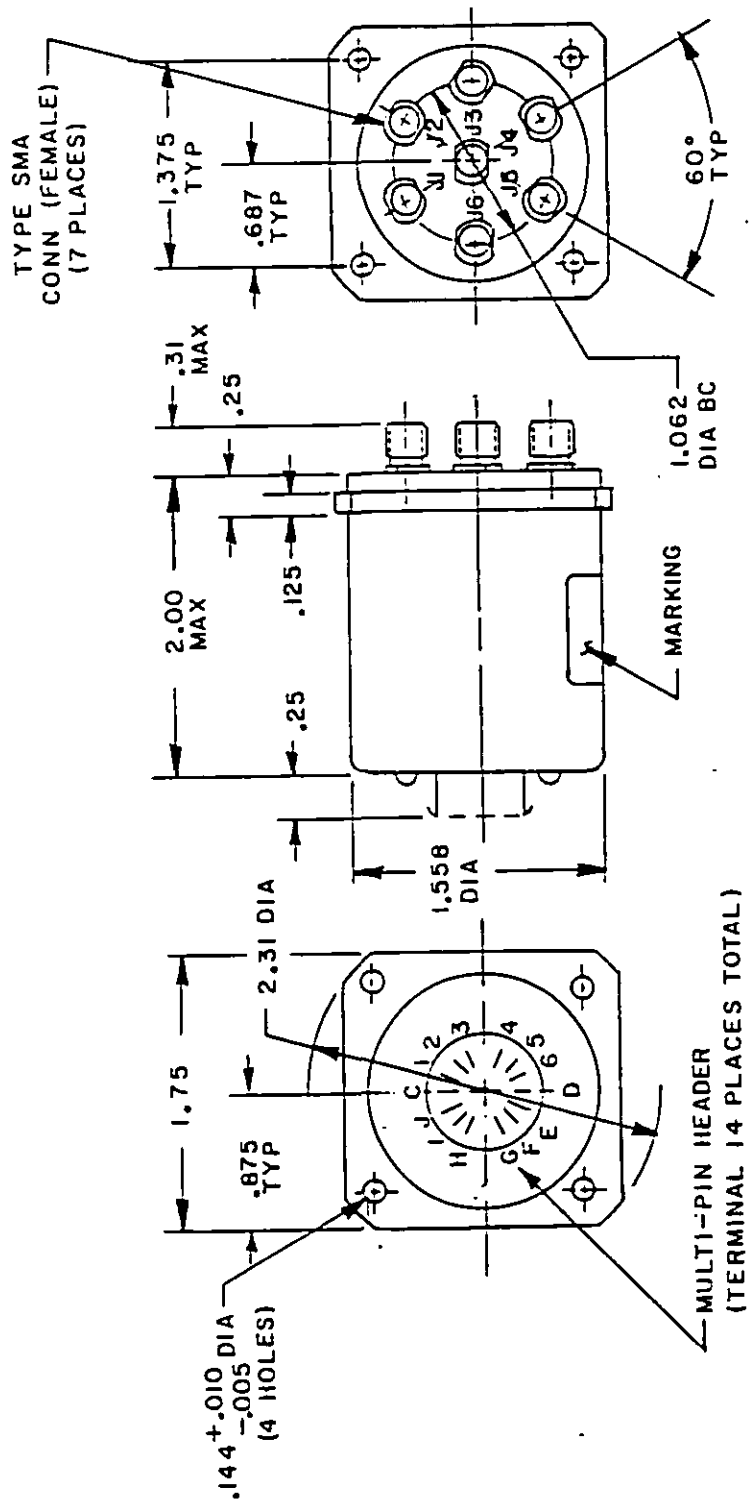
Inches	mm
.005	0.13
.010	0.25
.17	4.32
.177	4.50
.27	6.86
.50	12.70
1.60	40.64
1.800	45.72
2.00	50.80
2.25	57.15
2.43	61.72

SCHEMATIC NORMALLY OPEN WITH 50 OHM TERMINATIONS SHOWN IN POSITION I

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.010 (0.25 mm) for three place decimals and ±.03 (0.8 mm) for two place decimals.

Ⓑ FIGURE 6. Switch configuration and schematic, part number M3928/18-06.

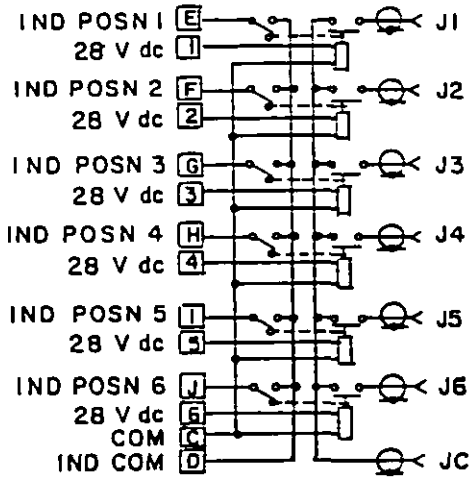
Source: https://assist.dla.mil -- Downloaded: 2026-10-16T22:19Z
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Inches	mm	Inches	mm
.005	0.13	.875	22.22
.010	0.25	1.062	26.97
.125	3.18	1.375	34.92
.144	3.66	1.558	39.57
.25	6.4	1.75	44.4
.31	7.9	2.00	50.8
.687	17.45	2.31	58.7

Ⓑ FIGURE 7. Switch configuration and schematic, part number M3928/18-07.

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SCHEMATIC (SHOWN IN DE-ENERGIZED POSITION)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.010 (0.25 mm) for three place decimals and ±.03 (0.8 mm) for two place decimals.

(B) FIGURE 7. Switch configuration and schematic, part number M3928/18-07 - Continued.

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(B) TABLE I. Electrical and performance characteristics.

Part no. M3928/18-	Figure no.	Housing	Frequency range dc to GHz	VSWR	Insertion loss (dB)	Isolation (dB)	Switch time (ms)	Position indication circuit and rating	Life cycles x 1000	Fail-safe or latching	Operating current (A) 1/	Holding current (A) 1/	Nominal operating voltage	Pickup voltage (less than)	Dropout voltage (less than)	Power and indicator connector	Weight (oz)
				Max	Max	Min	Max				Max	Max					Max
01N,S	1	I	18	1.5:1	0.5	60	20	NONE	1000	-	.17	.17	28 V dc	20 V dc	16 V dc	RIKQ7-8-7P (Deutsch) or equal	6
02N,S	2	I	18	1.5:1	0.5	60	20	NONE	1000	-	.17	.17	28 V dc	20 V dc	16 V dc	Solder terminals	6
03N,S	3	I	18 @ 2-7 @ 2-3	1.5:1 1.25:1	0.5 0.3	60 2/	50	NONE	1000	-	.22	.22	28 V dc	24 V dc	18 V dc 3/	Solder terminals	4.5
04N,S	4	I	18 @ 6-12 @ DC-6	1.5:1 1.4:1 1.25:1	0.5 0.4 0.25	60 60 70	20	NONE	1000	-	.17	.17	28 V dc	20 V dc	16 V dc	Solder terminals	6
4/ 05N,S	5	I	18 @ 6-12 @ DC-6	1.5:1 1.4:1 1.25:1	0.5 0.4 0.2	60 60 70	20	NONE	1000	-	.16	.16	28 V dc	20 V dc	16 V dc	MS3112E14-18PN	4
5/ 06N,S	6	I	18	1.5:1	0.5	60	15	2A@28 V dc	1000	-	.16	.16	28 V dc	20 V dc	16 V dc	MS3112E14-15P	6
07N,S	7	I	18	1.5:1	0.5	60	20	2A@28 V dc	1000	-	.18	.18	28 V dc	20 V dc	16 V dc	Solder terminals	6
08N,S	2	I	18.5	1.5:1	0.6	60	20	NONE	1000	-	.17	.17	28 V dc	20 V dc	16 V dc	Solder terminals	6

1/ At 28 V dc and 20°C.

2/ Isolation between connected ports and open ports shall be 6 dB minimum.

3/ Switch must return to deenergized position at no less than 2 volts.

4/ Contain ARC suppression diodes.

5/ Off-circuit output terminals are terminated with a 50 ohm, 1/2 watt resistor.

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REQUIREMENTS:

Dimensions and configurations: See figures 1 through 7.

Termination: Open, except for dash number 06 which is terminated.

Nominal impedance: 50 ohms.

RF connectors: Female connectors (7 places) shall meet the requirements of MIL-C-39012 and shall mate with SMA type male connectors per MIL-C-39012/55.

Electrical and performance characteristics: See table I.

Auxiliary contacts: 2 amps at 28 V dc for dash number 03.

RF power handling capability (average): 25 watts (minimum) except for dash number 03 which is 30 watts.

Transient interference (RFI): Required for M3928/18-03.

Vibration: Method I.

Operating temperature: -55°C to +85°C except for dash number 03 which is -55°C to +115°C.

Part number: M3928/18- (dash number from table I).

Substitutability: Switches covered by this specification sheet are substitutable for the manufacturer's or original equipment manufacturer's (OEM) part numbers specified in table II. This information in no way implies that the manufacturer's or OEM part number or NSN is suitable as a substitute for the military part number.

TABLE III. Substitution data.

Part number M3928/18-	OEM or manufacturer's code	OEM or manufacturer's part number	NSN 5985-
01	82152	146C70100	00-172-8187
02	82152	146C70600	00-005-2503
02	91417	118212-001	00-005-2503
03	97942	582R040H02	---
03	96341	MA7533-S009	---
04	94117	290099P007	01-133-4047
04	24022	CS-38S16-12	---
05	97871	339-0073-001	---
05	24022	CS-38S-16-14	---
06	20418	ES6396-01	01-234-8457
06	50667	M6423H30FT	---
07	20418	ES6395-01	
07	82152	146C71300	01-150-1270
08	12436	6010201-003	---
08	82152	146C70600-8	---

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CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - MI
Navy - OS
Air Force - 11, 99
DLA - ES

User activities:
Army - AV
Navy - AS, CG, MC
Air Force - 19

Preparing activity:
Navy - EC

Agent:
DLA - ES

(Project 5985-1016-10)

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